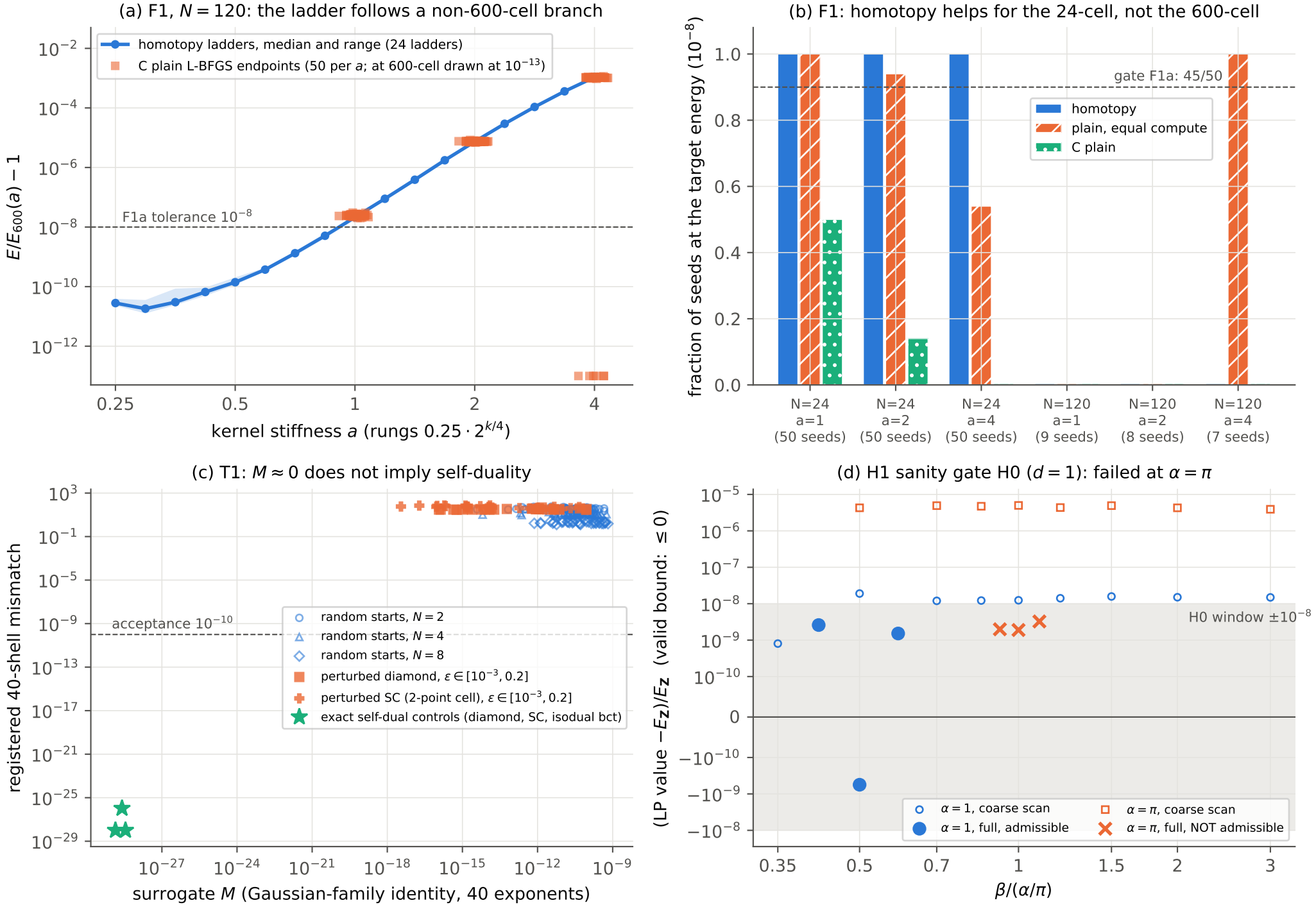

(a) F1, $N = 120$: the ladder follows a non-600-cell branch
homotopy ladders, median and range (24 ladders)
C plain L-BFGS endpoints (50 per a; at 600-cell drawn at 10^{-13})
F1a tolerance 10^{-8}
$E/E_{600}(a) - 1$
kernel stiffness a (rungs $0.25 \cdot 2^{k/4}$)
(b) F1: homotopy helps for the 24-cell, not the 600-cell
gate F1a: 45/50
homotopy
plain, equal compute
C plain
fraction of seeds at the target energy (10^{-8})
N=24 a=1 (50 seeds)
N=24 a=2 (50 seeds)
N=24 a=4 (50 seeds)
N=120 a=1 (9 seeds)
N=120 a=2 (8 seeds)
N=120 a=4 (7 seeds)
(c) T1: $M \approx 0$ does not imply self-duality
acceptance 10^{-10}
random starts, $N = 2$
random starts, $N = 4$
random starts, $N = 8$
perturbed diamond, $\varepsilon \in [10^{-3}, 0.2]$
perturbed SC (2-point cell), $\varepsilon \in [10^{-3}, 0.2]$
exact self-dual controls (diamond, SC, isodual bct)
registered 40-shell mismatch
surrogate M (Gaussian-family identity, 40 exponents)
(d) H1 sanity gate H0 ($d = 1$): failed at $\alpha = \pi$
H0 window $\pm 10^{-8}$
$\alpha = 1$, coarse scan
$\alpha = 1$, full, admissible
$\alpha = \pi$, coarse scan
$\alpha = \pi$, full, NOT admissible
(LP value $-E_Z$)/E_Z (valid bound: ≤ 0)
$\beta/(\alpha/\pi)$